ASHRAY SAXENA

Curriculum Vitae

Aachen, Germany | aashraysaxena@gmail.com | +49 152 2873 2051 | +91 81781 21793
i4mgr0ot.github.io/ashraysaxena.com | i4mGr0ot | linkedin.com/in/ashraysaxena | ORCID 0009-0001-8570-9160

RESEARCH INTERESTS

State estimation and navigation for aerial robots when GNSS and vision degrade; estimator consistency and uncertainty quantification; guidance, control and safe autonomy; aerial manipulation.

EDUCATION

Birla Institute of Technology and Science (BITS) Pilani, Pilani, India — Aug 2023 – Jul 2027 (expected)
B.E. (Hons.) Mechanical Engineering — CGPA **8.48/10** (US GPA 3.6/4.0)

RWTH Aachen University, Aachen, Germany — Aug 2026 – Feb 2027
International visiting student, Institute of Aerospace Systems (ILR). Bachelor's thesis within Project ATLAS, a joint project with Airbus and DLR funded by the German government.

RESEARCH EXPERIENCE

RWTH Aachen University, Institute of Aerospace Systems (ILR) Project ATLAS — Aug 2026 – Feb 2027
Supervisor: Univ.-Prof. Dr.-Ing. Eike Stumpf · Consultant: Aswin Thagadi Seenu, M.Sc.

- Thesis: *Credibility Estimation of Future Aircraft Concepts*. Developing a method, based on Dempster–Shafer evidence theory, to judge how far the design predictions for aircraft entering service in 2040–45 can be trusted when much of the input data is uncertain or comes from expert judgement.
- Validating the method on historical aircraft and test cases, then implementing it in UNICADO, the institute's aircraft design software, for the kerosene and hydrogen aircraft concepts studied in Project ATLAS.

University of Manchester · Visiting Research Intern — Jul 2026 – Oct 2026
Mechanical, Aerospace & Civil Engineering · Advisor: Dr. Kuppuraj Rajamanickam

- Designed a fuel atomiser for high-speed propulsion, studying how the injector shape affects the way the spray breaks up, to get more even mixing and steadier combustion.

LAT Aerospace · Research & Development Intern — Aug 2025 – Aug 2026
Aeromodelling, engine controls, and perception & autonomy teams · Supervisor: Priyansh Singh, Team Lead

- **IRIS navigation simulator.** Built IRIS, a simulator for testing how well a drone can navigate with only a camera and an IMU when GPS is unavailable. It generates realistic flights with photoreal camera images (Unreal Engine and Cesium) and IMU noise matched to a real sensor, runs our OpenVINS-based estimator, and checks both its drift and whether its reported uncertainty can be trusted. Drift stayed between 0.12 and 0.2 m over a 60 m test trajectory.
- **Flight validation.** Validated IRIS against 11 self-flown flight tests. When the estimator failed after a sharp turn on one flight, ran a walking test with the camera-IMU rig before and after every following flight and traced the failure to an inaccurate model of the time offset and mounting position between the camera and the IMU.
- **Square-root VIO.** Worked out how a square-root formulation (√VINS) could replace the filter backend in OpenVINS for better numerical stability, and built a low-cost camera-IMU rig with hardware timestamping to compare the two.
- **Engine control.** DOI Designed the electronic engine controller (FADEC) for the company's micro-turbojet. One processor runs the control law while a second, independent processor watches the engine limits and can cut the fuel if something goes wrong. Tested it against every failure case on a software model of the engine, and GTRE (India's Gas Turbine Research Establishment) certified the board for further deployment.
- **Blown-wing STOL study.** Compared two blown-wing layouts for the company's STOL aircraft, counter-circulation and distributed propulsion, by how much each delays stall and adds lift at low speed.
- **SkyNET high-lift framework.** Extended Prof. Drela's airfoil design codes (MSES, XFOIL, LINDOP) to propeller-blown 3D wings, with flaps and slats in every combination and their full deployment motion, so the team could compare high-lift designs much faster than with CFD. Also worked on the composite structure of the first uSTOL prototype.

BITS Pilani, ISHRAE · Undergraduate Researcher Jan 2026 – Aug 2026

Continuous-wave rotor pressure exchanger for supercritical CO_2 cycles · Advisor: Dr. Mani Sankar Dasgupta

- Experimental study of a pressure exchanger that controls the pressure waves inside its rotor channels; it reached 49% higher isentropic efficiency than conventional rotary-blade designs.

PUBLICATIONS

Accepted and published

[1] **A. Saxena** and S. Shrivastava, "How the elastic axis folds the transonic flutter boundary: shock energy transfer and conformally calibrated active learning," *Transport Phenomena* (De Gruyter Brill), accepted. arXiv May – Nov 2026

Shows how moving the elastic axis reshapes the transonic flutter boundary of a pitching and plunging wing section, and explains the resulting fold through energy transfer from shock motion, using SU2 Euler simulations and an actively learned, conformally calibrated surrogate.

[2] **A. Saxena** and S. Shrivastava, "RANS-verified surrogate design of wing sections for a uSTOL aircraft," *IAAI-27, Innovative Applications of Artificial Intelligence*, accepted. DOI 2026

Trains a surrogate on CFD results to search for wing sections that meet design targets, then checks the final designs with RANS CFD: across a tenfold range of Reynolds number and two turbulence models, predicted drag and lift held within 2.3% and 2.4%.

[3] V. Shah, **A. Saxena** and M. S. Dasgupta, "A 3D-printed cam-switch mechanism for verified steering mode transitions in a four-wheel independent steer-by-wire rover," *ACDSA 2027*, Brazil, accepted. Apr – Nov 2026

A low-cost 3D-printed rotary cam, with an encoder confirming each switch, that lets a four-wheel independently steered rover change reliably between turning on the spot and Ackermann steering; tested on the CRISS autonomous rover.

[4] **A. Saxena**, "Rapid aerodynamic optimization of aerospace structures: engine nacelles," *Acceleron Aerospace Journal*, vol. 7, no. 1, pp. 1991–2003, Sep. 2026. DOI 2026

A fast Python framework for shaping engine nacelles with an in-house viscous-potential flow solver, showing that local reshaping of the nacelle lip can narrow the design space before expensive Navier–Stokes simulations.

[5] **A. Saxena**, "Modular end effector architecture for autonomous environmental sensing," *Global Scientific Journal*, vol. 13, no. 10, pp. 1726–1731, Oct. 2025. PDF 2025

A modular robot end effector combining a gripper, soil sampler, microscope, miniature spectrometer and environmental sensors in one unit, which switches between inspection, scanning and analysis on its own.

Under review

[6] **A. Saxena**, "A critical-layer instability of structural-acoustic wall modes in a two-flexible-plate channel with Poiseuille flow: closed-form low-Mach corrections and parity-protected absorption," *Journal of Fluid Mechanics*, under review. arXiv DOI 2026

Identifies a new convective instability of the wall mode in a flexible-walled channel with Poiseuille flow, driven by the critical layer; it grows with wall flexibility and vanishes for rigid walls. Also derives closed-form low-Mach corrections for all coupled modes.

[7] **A. Saxena**, "Joint optimisation of active aerodynamics and electric energy deployment for minimum lap time in 2026 Formula One," *Proc. IMechE Part D: Journal of Automobile Engineering*, under review. DOI 2026

Finds the fastest 2026 Formula One qualifying lap by optimising the use of movable aerodynamics and electrical energy within the new regulations; once calibrated, predicted lap times match real 2026 pole laps within 0.17%.

Conference abstract

[8] **A. Saxena**, "Frequency-domain analysis of thermoacoustic coupling in liquid-propellant rocket engines," *IEEE Aerospace Conference 2026*, accepted abstract. 2026

A COMSOL frequency-domain model of how acoustic waves couple with heat release in a rocket combustion chamber, used to screen chamber shapes and damping early.

PROJECTS AND COMPETITIONS

CRISS Robotics, BITS Pilani · Team Captain, space robotics and rover team Jan 2024 – present

- Led 50+ students across four subsystems on a $20,000+ budget. Placed 3rd at the European Rover Challenge 2025 and became the first Indian team to qualify for the Australian Rover Challenge 2026.
- Designed the rover's chassis (machined and welded aluminium), wheels, actuators, steering and robotic arm. The steering can turn the rover on the spot or use Ackermann geometry, switched by the cam mechanism in [3].
- Cut the rover's mass from 78 kg to 51 kg by 3D printing about half of the connecting parts and laser-cutting the covers.
- Built a camera-based autonomous spectroscope with ROS and OpenCV, and designed the imaging and infrared spectrometer payloads and the soil sample collection system.

M-SWA: Multi-Modal Semantic Workbench Assistant · SIRE grant, BITS Pilani Sep 2026 – present

- A robot arm you can talk to, in development: a vision-language model finds the object you mean, and the arm points at it with a laser, hands it over and remembers it. Heat and sound sensors answer questions such as whether a motor is overheating.

Czerny–Turner Raman Spectrophotometer · SOLVE grant, BITS Pilani May 2024 – Nov 2024
Advisor: Dr. Surojit Pande, BITS Pilani

- Built a Raman spectrophotometer that detects organic biosignatures without damaging the sample.

HONOURS AND AWARDS

European Rover Challenge, 3rd place, CRISS Robotics 2025
Australian Rover Challenge, first Indian team to qualify, CRISS Robotics 2026
International Rover Design Challenge (SPROS), 3rd place, CRISS Robotics 2025
Project Prototype Presentation, APOGEE, BITS Pilani, 1st place (twice) 2024, 2025
3rd International Workshop on GESD, BITS Pilani, 2nd place for poster presentation 2024
KVPY Scholar, All-India Rank 189 among 200,000+ candidates; full marks in Mathematics and Physics 2022
NTSE Scholar, All-India Rank 4 among 350,000+ candidates; full marks in Mental Ability 2021
Mathematics Olympiad (IOQM), All-India top 100 for three consecutive years 2021 – 2023

TEACHING AND MENTORING

Undergraduate Teaching Assistant, Department of Mechanical Engineering, BITS Pilani Aug 2025 – present
Computer-Aided Design, ME F381 (Aug 2026 – present); Quality Control, Assurance and Reliability, ME F434 (Jan – May 2026); Mechanics of Solids, ME F211 (Aug – Dec 2025).

Writing Coach, RISE Global Aug 2025 – Mar 2026
Mentored 12+ high school students through writing and publishing their first research papers.

LEADERSHIP AND ENTREPRENEURSHIP

CTO & Co-Founder, AtmosClear Pvt Ltd Dec 2024 – Feb 2026
Built SAMAP, a self-navigating mobile air purifier. Raised $32,742 for 2% equity from NIDHI PRAYAS (Government of India) and Forth Tech Limited; later exited.

Team Manager, Inspired Karters Gravity (karting and Formula Student), BITS Pilani Aug 2024 – Aug 2025
Managed 40+ students and $10,000 in funding; designed the Formula Student car with regenerative braking.

EARLIER ENGINEERING ROLES

India Space Lab, Satellite Communication Intern: CubeSat radio link and ground control Jul – Aug 2025
DHIO Research & Engineering, CFD Intern: rocket combustion instability [8] May – Jul 2025
CrazyGoldFish.ai, Robotics Development Intern: page-turning book scanner with OCR Oct 2024 – Feb 2025

TECHNICAL SKILLS

Estimation	MSCKF/VIO (OpenVINS), square-root filtering, NEES consistency, ROS2, OpenCV
Optimisation, UQ	CasADi/IPOPT, NSGA-II, Dempster–Shafer theory, Gaussian processes, conformal prediction
Aero, simulation	SU2, ANSYS Fluent, OpenFOAM, COMSOL, MSES/XFOIL, OpenVSP, UNICADO
Programming	C++, C, Python, MATLAB/Simulink, PyTorch, Git; STM32, ESP32, Altium, KiCad
Hardware, CAD	UAV flight testing, wind tunnels, CNC, 3D printing; SolidWorks, CATIA V5, NX, Fusion 360